REPORT ON MINI PROJECT

Name: Masthabanu S

Batch Code: TN_DA_FNB10

Email ID: masthas26@gmail.com

Contact No: 6369490423

Project Title: Cardiovascular Health Risk Analysis Using Clinical and Lifestyle Data

Project Domain: Healthcare

Tools Used: Excel & Power BI

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Mentor Name: Kumaran M

Raw Dataset Link:
https://docs.google.com/spreadsheets/d/1gC5VRfR5srvqGZ_MkyCb_hAVk4NPd1QO/edit?usp=sharing&ouid=117042592191127031459&rtpof=true&sd=true

Cleaned Dataset Link:
https://docs.google.com/spreadsheets/d/1G5_onNkPizTFT1xaPj5xHmZ7zb5ijnME/edit?usp=sharing&ouid=117042592191127031459&rtpof=true&sd=true

Cardiovascular Health Risk Analysis Using Clinical and Lifestyle Data

This project aims to analyze and visualize cardiovascular health data to identify risk patterns and health trends. The dataset contains multiple clinical and lifestyle indicators that influence heart disease risk.The dataset includes key patient health attributes such as age, gender, body measurements, cholesterol levels, blood sugar levels, blood pressure readings, lifestyle habits, and physical activity levels.The objective is to transform raw health data into meaningful insights using data cleaning, feature engineering, and visualization techniques.

To enhance the analysis, several derived columns were calculated, including:

- **BMI Level** – Classified as Underweight, Normal, Overweight, and Obese based on BMI values to assess obesity-related risk.
- **Blood Pressure Category** – Grouped into Normal, Elevated, and Hypertension levels to evaluate the impact of blood pressure on heart disease.
- **Fasting Blood Sugar Category** – Grouped into Normal, Prediabetes, and Diabetic categories according to fasting blood sugar measurements to evaluate its influence on heart disease risk.
- **CVD Risk Level** – Classified as Low, Moderate, and High based on combined clinical indicators such as age, blood pressure, cholesterol, BMI, and fasting blood sugar to determine the overall cardiovascular disease risk of each patient.

The analysis and visualization help to answer questions like:

- How does BMI distribution vary between Male and Female patients?
- Is the average cholesterol level within the recommended healthy range?
- Which BMI category shows the highest percentage of high cardiovascular risk?
- How many individuals fall under low and high CVD risk categories?
- How many individuals fall under normal, prediabetic, and diabetic categories?
- How does cardiovascular risk vary across different blood pressure categories?
- To what extent does total cholesterol level influence the cardiovascular risk score?

Tools & Technologies used:

- Microsoft Excel - Data Cleaning, Transformation and formula based calculation.
- Power BI - Interactive dashboards and Visual analytics
- Data Visualization Technique - KPI Cards, Gauge, Clustered column chart, 100% Stacked Column chart, Scatter chart, Donut chart, Clustered Bar chart.

DATA CLEANING:

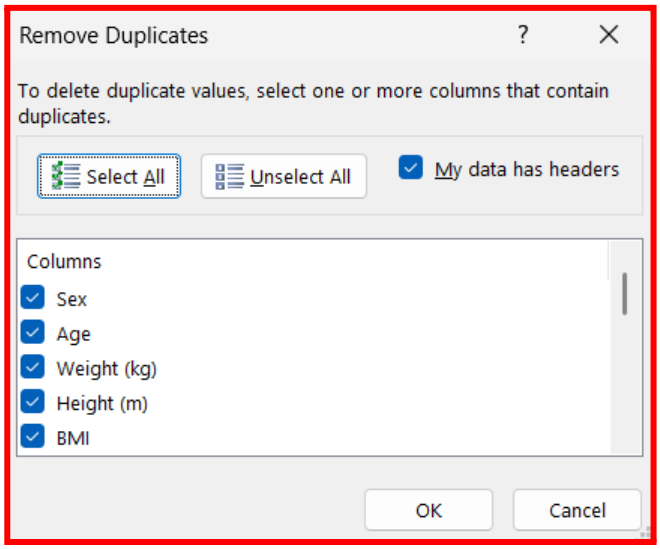
1. **INCONSISTENCY IN CVD RISK LEVEL**

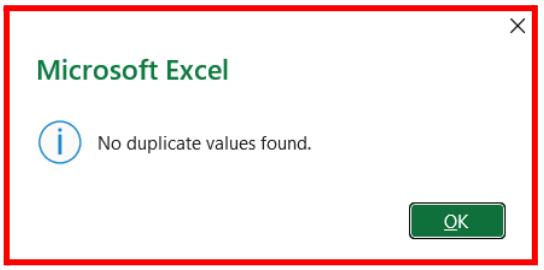
   Inconsistent text formatting were identified (values such as HIGH, LOW, INTERMEDIARY were in uppercase). These were standardized into a consistent format (High, Low, Intermediary) to ensure uniformity and improve data clarity.

```
=PROPER([@[CVD Risk Level]])
```

AJ	AK
CVD Risk Level	CVD Risk Level2
INTERMEDIARY	Intermediary
HIGH	High
INTERMEDIARY	Intermediary
INTERMEDIARY	Intermediary
INTERMEDIARY	Intermediary
HIGH	High
INTERMEDIARY	Intermediary
HIGH	High
INTERMEDIARY	Intermediary

2. **REMOVING DUPLICATES**





3. **MISSING VALUES IN AGE**

Missing values were identified in the Age column and replaced using the average (mean) age of the dataset to maintain consistency and avoid data loss.

```
=IF(B2="",AVERAGE($B$2:$B$1530),B2)
```

B
Age
32
55
44
32
58
33

C
Age2
32
55
47
44
32
58
33

4. **MISSING VALUES IN WEIGHT(Kg)**

Missing values in the Weight column were calculated using the BMI formula (Weight = BMI × Height²) wherever BMI and Height values were available, ensuring accurate data completion for further analysis.

```
=IF(D2="", H2*(F2^2),D2)
```

D
Weight (kg)
69.1
118.7
108.3
99.5
117.9
117.9

E
Weight(Kg)
69
119
90
108
100
118
118

5. **MISSING VALUES IN HEIGHT(m)**

Missing values in the Height column were calculated using the BMI formula (Height = √(Weight / BMI)) wherever BMI and Weight values were available, ensuring consistency and accuracy in the dataset.

```
=IF(F2="", SQRT(E2 / H2),F2)
```

F	G
Height (m)	Height(m)
1.71	1.71
1.69	1.69
1.8	1.80
1.8	1.80
	1.90
1.71	1.71
	1.73
	1.78

6. **MISSING VALUES IN BMI**

The BMI column contained missing values; therefore, BMI was recalculated using the standard formula (Weight in kg / Height in meters²)

```
= IF(H2="",E2 / (G2^2),H2)
```

H	I
BMI	BMI2
	37.3
	16.8
	24.3
	25.2
15.4	15.4
15.4	15.4
	30.4
	32.4
	21.7
	29.0

7. MISSING VALUES IN WAIST-TO-HEIGHT RATIO

Missing values were calculated using abdominal circumference and height through a conditional formula to ensure data completeness and accurate assessment of central obesity.

(Waist-to-HeightRatio(L2)= AbdominalCircumference(K2)÷(Height(g2)*100))

```
=IF(L2="",K2/(G2*100),L2)
```

L
Waist-to-Height Ratio
0.504
0.548
0.504
0.55

M
Waist-to-Height Ratio2
0.504
0.459
0.544
0.501
0.548
0.504
0.495
0.572
0.550

8. MISSING VALUES IN ABDOMINAL CIRCUMFERENCE

Calculated missing values using the formula: Abdominal Circumference = Waist-to-Height Ratio × (Height*100), to maintain data accuracy and support further analysis.

```
=IF(K2="",M2*(G2*100),K2)
```

K
Abdominal Circumference (cm)
70
73.4
70.2

N
Abdominal Circumference(cm)
109.8
79.6
70.0
73.4
87.5
92.6
70.2
103.3
83.2
106.4

9. MISSING VALUES IN HDL(mg/DL)

Missing values were filled using the average HDL value to preserve data consistency for cardiovascular analysis.

```
=IF(P2="",AVERAGE($P$2:$P$1530),P2)
```

P	Q
HDL (mg/dL)	HDL (mg/dL)2
67	67
72	72
71	71
	56
	56
67	67
71	71
	56
67	67

10. MISSING VALUES IN ESTIMATED LDL(mg/DL)

Missing and inconsistent values, including negative entries, were corrected by replacing them with the median LDL value, ensuring the dataset remained accurate for cardiovascular risk analysis.

```
=IF(R2<=0,MEDIAN($R$2:$R$1530),R2)
```

R	S
Estimated LDL (mg/dL)	Estimated LDL (mg/dL)2
0	109
7	7
	109
	109
	109
-1	109
1	1
	109
-2	109

11. MISSING VALUES IN TOTAL CHOLESTEROL

To calculate cholesterol the formula is:

Cholesterol=Estimated LDL+HDL+(Triglycerides/5)

Here, Triglycerides is not present, So I Calculated Triglycerides by:

Triglycerides=5×(Cholesterol+Estimated LDL+HDL)

```
=IF(O2="",[@[Estimated LDL (mg/dL)2]]+[@[HDL (mg/dL)2]]+([@Triglycerides]/5),O2)
```

O
Total Cholesterol (mg/dL)
248
250
247

U
Total Cholesterol(mg/dL)
248
516
396
284
206
250
247
234
282
516

12. MISSING VALUES IN FASTING BLOOD SUGAR (mg/DL)

Missing values in the Fasting Blood Sugar column were replaced with the average (mean) fasting blood sugar of the dataset.

```
=IF(V2="",AVERAGE($V$2:$V$1530),V2)
```

V
Fasting Blood Sugar (mg/dL)
111
142
109
145
88
130

W
Fasting Blood Sugar (mg/dL)2
111
142
117
109
145
88
130

13. MISSING VALUES IN SYSTOLIC BP

The Blood Pressure column contained values in the format of Systolic/Diastolic. The Systolic BP was extracted using the following Excel formula:

```
=LEFT(AB2,FIND("/",AB2)-1)
```

AB	AD	AC
Blood Pressure (mmHg)	Systolic BP2	Systolic BP
125/79	125	125
139/70	139	139
104/77	104	104
140/83	140	140
144/83	144	144
142/90	142	142
96/63	96	96
115/73	115	115
107/95	107	
133/83	133	133

14. MISSING VALUES IN DIASTOLIC BP

Similarly, the second number (after the slash) was extracted from Blood Pressure(mmHg) column to fill the Diastolic BP column.

```
=RIGHT(AB2,LEN(AB2)-FIND("/",AB2))
```

AB	AE	AF
Blood Pressure (mmHg)	Diastolic BP	Diastolic BP2
139/70	70	70
104/77	77	77
142/90	90	90
120/89	89	89
121/79		79
137/70	70	70
148/97		97
97/75	75	75
136/80		80
130/75	75	75

15. MISSING VALUES IN CVD RISK SCORE

The CVD Risk Score column quantifies cardiovascular risk. Missing values were handled by replacing them with the mean (average) CVD Risk Score of the dataset.

```
=IF(AH2="",AVERAGE($AH$2:$AH$1530),AH2)
```

AH	AI
CVD Risk Score	CVD Risk Score2
17.93	17.93
	16.96
17.76	17.76
17.73	17.73
14.90	14.90
20.32	20.32
21.03	21.03
17.17	17.17
17.52	17.52

CALCULATED COLUMN:

1. Created a new calculated column named BMI Level to categorize BMI as Underweight, Normal, Overweight, or Obese using the following formula:

```
=IF(I2<18.5,"Underweight",IF(I2<25,"Normal",IF(I2<30,"Overweight","Obese")))
```

I	J
BMI2	BMI Level
23.6	Normal
41.6	Obese
26.9	Overweight
33.4	Obese
28.8	Overweight
33.7	Obese
35.6	Obese
33.1	Obese
31.8	Obese
28.2	Overweight
29.5	Overweight
26.4	Overweight

2. Created a new calculated column named Fasting Blood Sugar Category to categorize fasting blood sugar as Normal, Diabetes, Prediabetes using the following Excel formula:

```
=IF(W2<100,"Normal",IF(W2<126,"Prediabetes","Diabetes"))
```

W	X
Fasting Blood Sugar (mg/dL)2	Fasting Blood Sugar Category
111	Prediabetes
135	Diabetes
114	Prediabetes
91	Normal
141	Diabetes
82	Normal
87	Normal
96	Normal
70	Normal
126	Diabetes
74	Normal
121	Prediabetes
97	Normal
122	Prediabetes

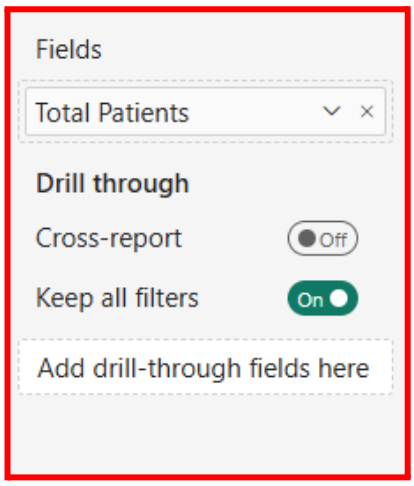
DATA VISUALIZATION:

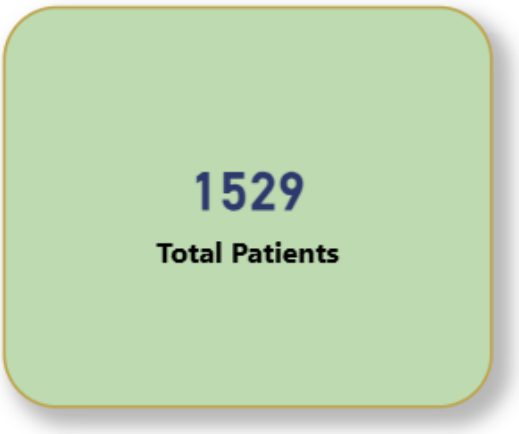
The visualization provides a comprehensive overview of how factors such as Body Mass Index (BMI), blood pressure, cholesterol levels, fasting blood sugar, smoking habits, and physical activity influence cardiovascular disease (CVD) risk.

- **Total Patients**

  The card shows the total count of patients whose clinical and lifestyle data were analyzed.Knowing the total number of patients helps in interpreting percentages, proportions, and comparisons shown in other visualizations.

```
Total Patients = COUNTA(Cleaned[Sex])
```



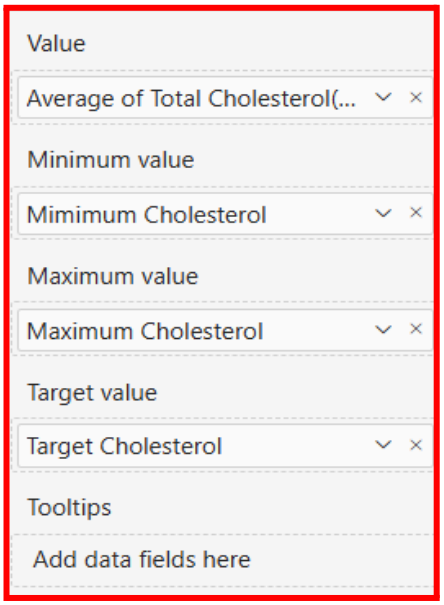


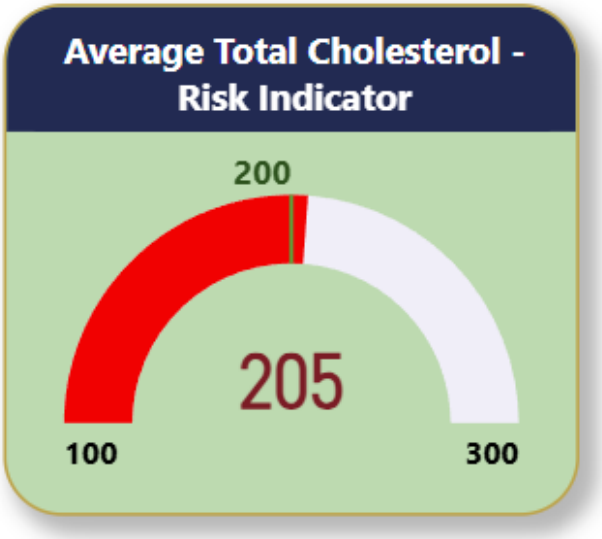
- **Average Total Cholesterol - Risk Indicator**
  This chart was created to assess whether the patient's average cholesterol level is within the healthy clinical range. Total cholesterol is a critical clinical indicator in assessing cardiovascular disease (CVD) risk. According to clinical guidelines, a total cholesterol level below 200 mg/dL is considered desirable, while higher values may indicate increased risk of heart disease.

```
Minimum Cholesterol = 100
```

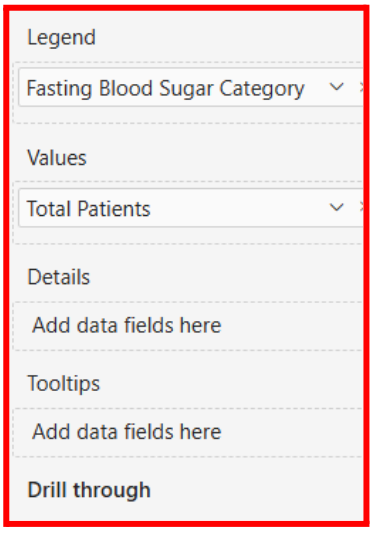
```
Maximum Cholesterol = 300
```

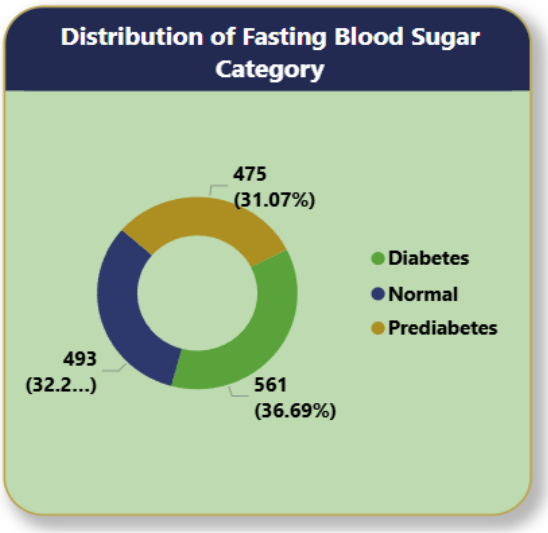
```
Target Cholesterol = 200
```





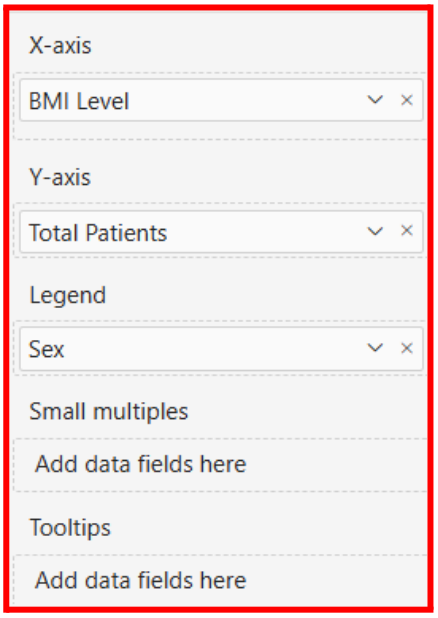
- **Distribution of Fasting Blood Sugar Category**
  This donut chart shows how patients are distributed among different fasting blood sugar categories such as Normal, Prediabetes, and Diabetes.The purpose of this chart is to understand how many individuals have normal blood sugar levels and how many are at risk due to elevated levels. Since high blood sugar is a major risk factor for cardiovascular disease, this visualization helps identify the proportion of patients who may require medical attention or lifestyle changes.

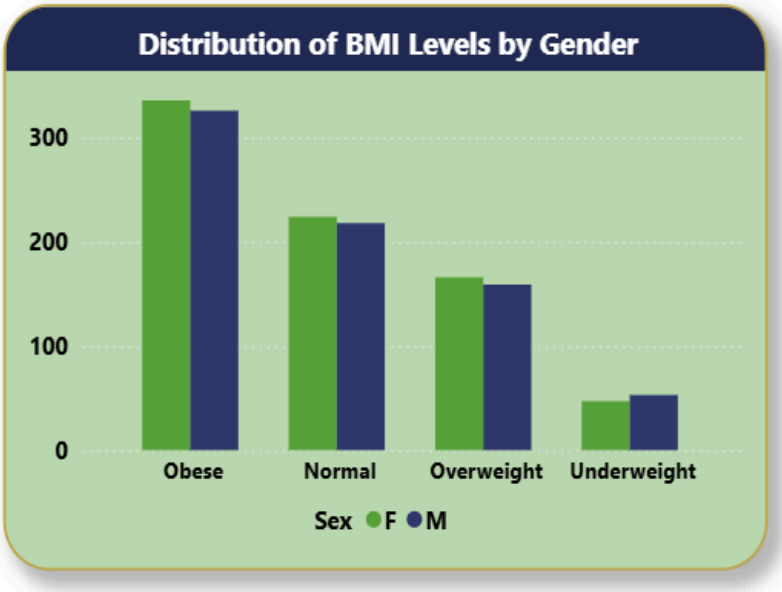




- **Distribution of BMI Level by Gender**

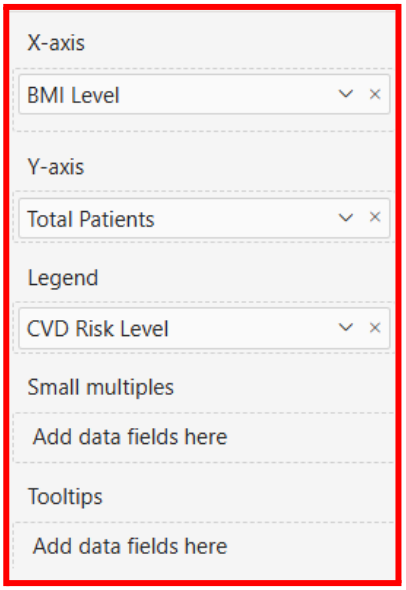
  The purpose of this chart is to identify whether there are noticeable differences in body weight patterns between males and females. By analyzing the distribution, it becomes easier to determine which gender has a higher proportion of overweight or obese individuals, which are important risk factors for cardiovascular disease.Understanding gender-based BMI differences helps in identifying vulnerable groups and supports targeted health interventions and preventive strategies.

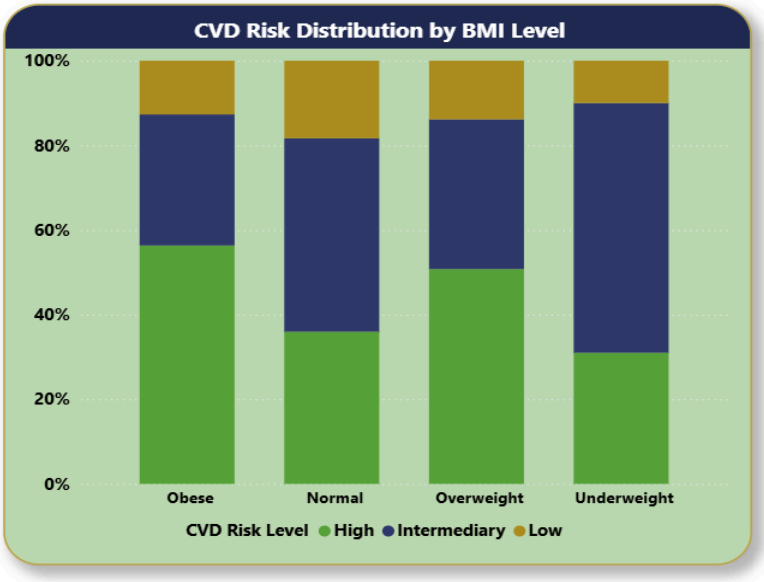




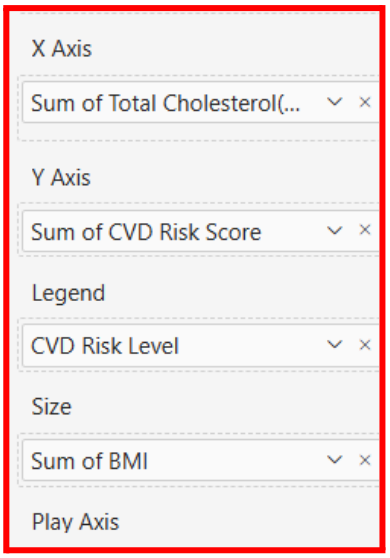
- **CVD Risk Distribution by CVD Level**

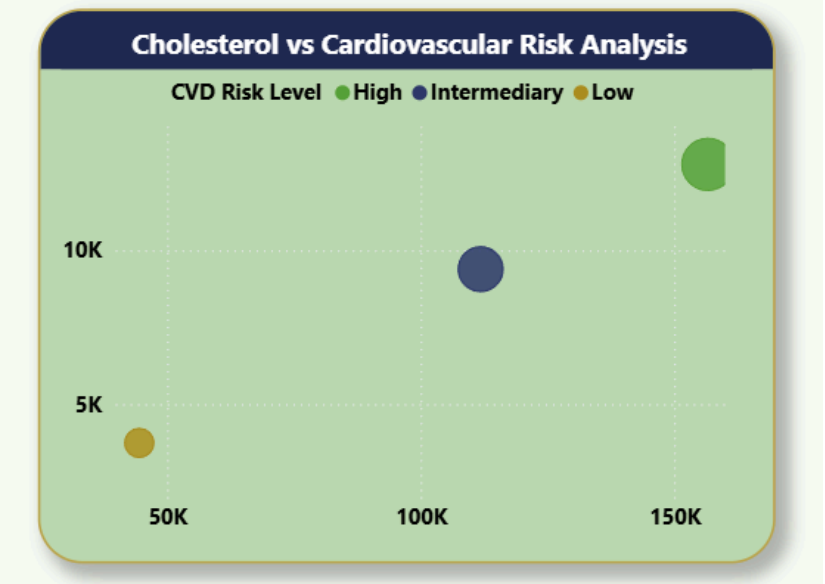
  This represents the proportional distribution of cardiovascular disease (CVD) risk levels within each BMI category. Instead of displaying raw counts, the chart shows the percentage composition of Low, Medium, and High risk individuals across Underweight, Normal, Overweight, and Obese groups. It becomes easier to identify trends such as a higher percentage of high-risk individuals within overweight or obese categories.





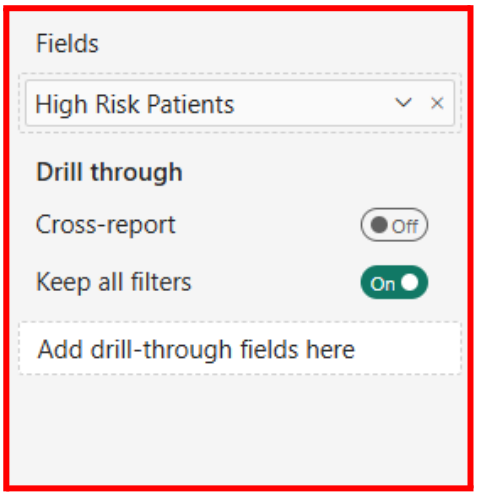
- **Cholesterol vs Cardiovascular Risk Analysis**
  This was created to examine the relationship between total cholesterol levels and cardiovascular disease (CVD) risk score. Unlike summary charts, this visualization displays individual data points, allowing a more detailed analysis of how cholesterol levels influence overall risk. The purpose is to identify whether higher cholesterol levels are associated with increased cardiovascular risk scores.



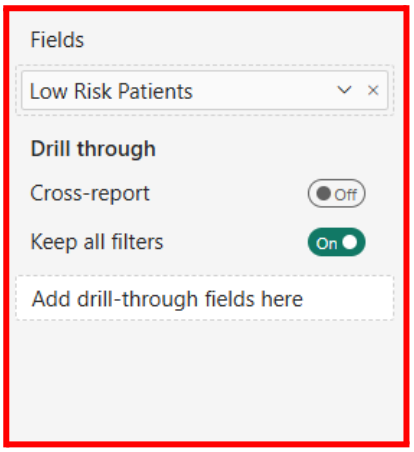


- **High Risk & Low Risk Patients**
  The KPI cards provide a high-level summary of the cardiovascular risk distribution within the dataset. These indicators display the number of patients categorized under Low Risk and High Risk groups. This provides a quick overview of the overall health risk distribution within the dataset.

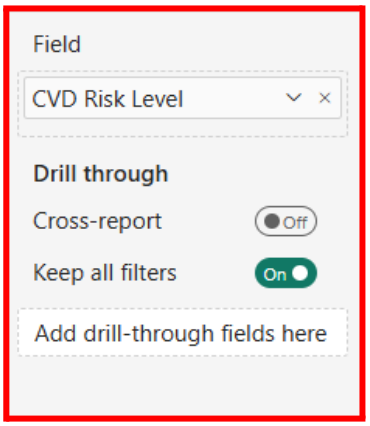
```
High Risk Patients = CALCULATE(COUNTROWS('Cleaned'),Cleaned[CVD Risk Level] = "High")
```

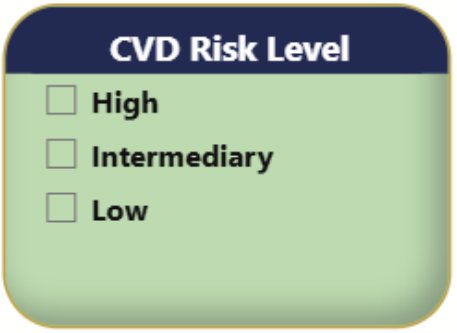


```
Low Risk Patients = CALCULATE(COUNTROWS(Cleaned),Cleaned[CVD Risk Level]="Low")
```

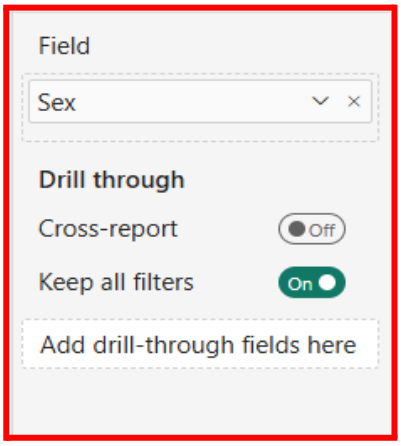


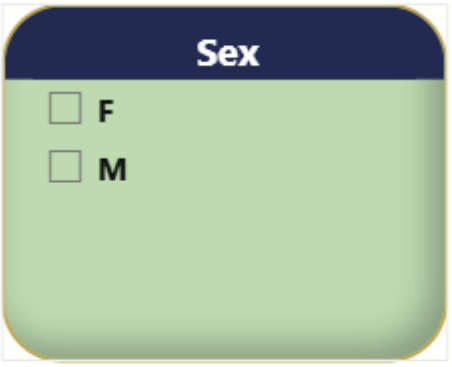
- **CVD Risk Level Slicer**
  The CVD Risk Level slicer was implemented to enable dynamic filtering of the dashboard based on cardiovascular risk categories (e.g., Low, Intermediary, High). This allows users to focus specifically on a particular risk group and analyze associated clinical and lifestyle characteristics.



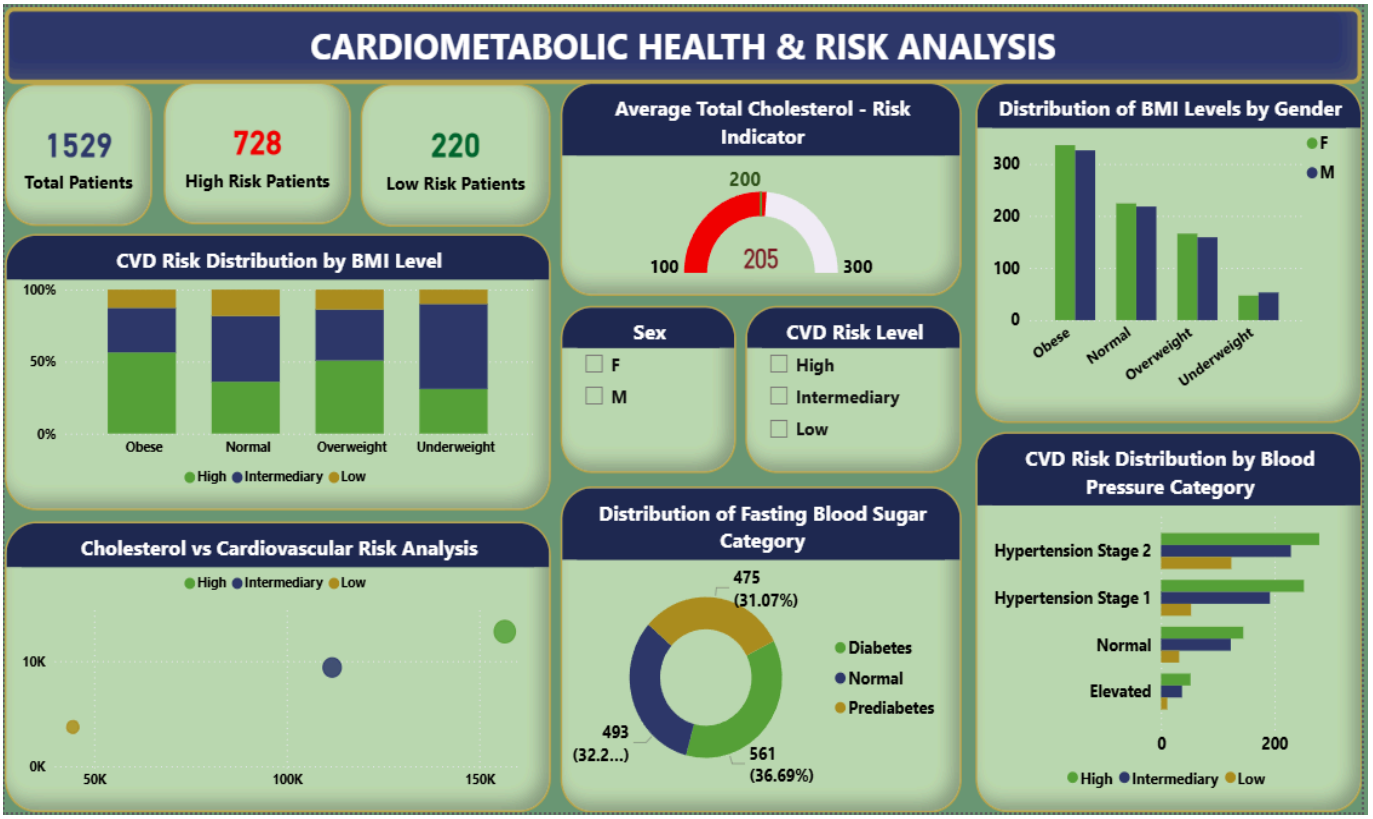


- **Sex Slicer**
  The Sex slicer was added to facilitate gender-based analysis within the dataset. It allows users to compare male and female populations across all health indicators and risk distributions.





DASHBOARD:



The Cardiometabolic Health & Risk Analysis provides a comprehensive analysis of cardiovascular risk by integrating key clinical indicators such as BMI, cholesterol, fasting blood sugar, and blood pressure. It combines summary cards, distribution charts, and relationship visualizations to evaluate how these factors influence CVD risk levels. The structured layout enables clear identification of high-risk groups and supports data-driven health assessment.

Key Features of Dashboard Include:

- **Risk Summary Indicators :** The dashboard includes KPI cards displaying Total Patients, Low Risk, and High Risk counts. These indicators provide an immediate understanding of the population size and overall cardiovascular risk distribution.
- **Cholesterol Assessment :** A gauge visualization presents the average total cholesterol level, enabling quick evaluation of whether the population's lipid levels fall within safe or elevated ranges.
- **BMI Distribution by Gender :** A clustered column chart compares BMI levels across male and female patients. This visualization highlights gender-based variations in body mass categories.

- **BMI Distribution by Gender :** A clustered column chart compares BMI levels across male and female patients. This visualization highlights gender-based variations in body mass categories.
- **BMI Influence on CVD Risk :** The 100% stacked bar chart illustrates the proportion of cardiovascular risk levels within each BMI category, demonstrating how increasing BMI may impact risk classification.
- **Cholesterol and Risk Score Relationship :** A scatter chart visualizes the relationship between total cholesterol and CVD risk score, categorized by risk level. This chart helps identify patterns and potential positive correlation between lipid levels and cardiovascular risk.
- **Fasting Blood Sugar Distribution :** A donut chart displays patient distribution across fasting blood sugar categories, showing the prevalence of normal, prediabetic, and diabetic conditions within the dataset.
- **Blood Pressure Impact on CVD Risk :** A clustered bar chart examines the relationship between blood pressure categories and cardiovascular risk levels, highlighting how hypertension stages correspond to increased risk.

OVERVIEW OF THE PROJECT :

Cardiometabolic Health & Risk Analysis

This dashboard provides a comprehensive analysis of 1,529 patients to understand cardiovascular disease (CVD) risk using key health indicators such as BMI, blood pressure, cholesterol, and blood sugar levels. The objective of the project is to identify high-risk patients and understand the major contributing factors to heart disease.

- **High Risk Population:** Nearly half of the patients fall under High CVD Risk. This indicates a serious health concern in the population and needs immediate lifestyle and medical intervention.
- **BMI Analysis:** Overweight and Obese patients show the highest CVD risk levels.Normal BMI patients mostly fall under low or intermediate risk. Increased body weight strongly contributes to cardiovascular risk.
- **Blood Pressure Impact:** Patients in Hypertension Stage 1 and Stage 2 show highest CVD risk. Elevated blood pressure directly increases heart complications. Uncontrolled hypertension is a major driving factor for CVD.

- **Cholesterol Levels:** The average cholesterol is 205 mg/dL, slightly above safe limit. High cholesterol patients are mostly in the High Risk category. This can block arteries and increase heart attack chances.
- **Diabetes & Prediabetes Concern**: Over 65% of patients are either diabetic or prediabetic. These groups show significantly higher CVD risk. Blood sugar imbalance significantly increases long-term heart risk.

This dashboard analyzes cardiovascular risk among patients using key health indicators like BMI, blood pressure, cholesterol, and blood sugar. The dashboard helps identify high-risk groups and shows how lifestyle-related factors contribute to heart disease.The insights support early diagnosis and preventive healthcare decisions.

The main causes of increasing heart risk in this population are obesity, high blood pressure, high cholesterol, and diabetes. These conditions are strongly connected, and without early control, they can lead to serious cardiovascular diseases.